

AQUÍ VA EL TÍTULO DEL ARTÍCULO

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ABSTRACT. Aquí va el resumen

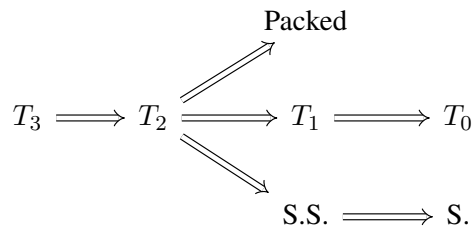
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1. INTRODUCTION

[PP21, SS06a, Hoc69]

[Sim06a, Sex03, Joh82]



$$\begin{array}{ccccc}
& & (\mathbf{H}) + (\mathbf{sfit}) & & \\
& \nearrow & & \searrow & \\
(\mathbf{reg}) & \xRightarrow{\quad} & (\mathbf{sH}) & \xRightarrow{\quad} & T_1 \\
& \searrow & & \nearrow & \\
& & (\mathbf{fit}) & \xRightarrow{\quad} & (\mathbf{sfit}) \\
T_2 & \xRightarrow{\quad} & \mathbf{Neat} & \xRightarrow{\quad} & T_1
\end{array}$$

$$\begin{array}{ccccc}
& & T_1 & & \\
& \nearrow & \uparrow & \nwarrow & \\
(\mathbf{H}) & \cdots \triangleright & KC & \cdots \triangleright & \mathbf{Neat} \\
& \searrow & \downarrow & \swarrow & \downarrow \\
& & \mathbf{S. S.} & \xRightarrow{\quad} & \mathbf{S.}
\end{array}$$

2. PRELIMINARIES

Aquí hay
que poner la
referencia.

Necesitamos quitar toda esta sección.

- Frames, nucleis, A_j
- u_a, v_a, w_a
- Points, $\mathcal{U}_*$, $\mathcal{U}$ spaciality
- Open filters,
- Fitted nucleis
- HM and HMJ

Let $f: A \rightarrow B$ be a frame morphism with right adjoint $f_*: B \rightarrow A$. If $F \subseteq A$ and $G \subseteq B$ are filters, we define:

$$(1) \quad b \in f^*F \iff f_*(b) \in F \quad a \in f_*G \iff f(a) \in G,$$

where f^*F and f_*G are filters in B and A , respectively.

Proposition 2.1. *For $f = f^*: A \rightarrow B$ a frame morphism and $G \in B^\wedge$, then $f_*G \in A^\wedge$.*

Proof. By (1), f_*G is a filter in A . We need that f_*G satisfy the open filter condition. Let $X \subseteq A$ be such that $\bigvee X \in f_*G$, with X directed. Then

$$Y = \{f(x) \mid x \in X\}$$

is directed and $f(\bigvee X) = \bigvee f[X] = \bigvee Y \in G$. Since G is an open filter and $f(\bigvee X) = \bigvee f[X] \in G$, there exists $x \in X$ such that $f(x) \in G$. Therefore, $x \in f_*G$. $\square$

3. COMPACT QUOTIENTS AND ADMISSIBILITY INTERVALS

- $A^\wedge \longleftrightarrow A_j$
- v_F, w_F admissibility intervals
-

4. SEPARATION PROPERTIES INDUCED BY COMPACT QUOTIENTS

4.1. Neat frames.

Definition 4.1. Neat definition

Proposition 4.2. Let (F, Q, G) be an HM triple, with

$$F \in A^\wedge \quad \text{and} \quad G = \nabla(Q) \in (\mathcal{O}S)^\wedge.$$

If

$$k \in [v_G, w_G],$$

then

$$\mathcal{U}_* \circ k \circ \mathcal{U} \in [v_F, w_F].$$

Proof. Since k is a nucleus on $\mathcal{O}S$, the composite

$$\mathcal{U}_* \circ k \circ \mathcal{U}$$

is a nucleus on A . Thus, it is enough to show that

$$\nabla(\mathcal{U}_* \circ k \circ \mathcal{U}) = F.$$

Let $x \in A$. Since (F, Q, G) is an HM triple,

$$x \in F \iff Q \subseteq \mathcal{U}(x) \iff \mathcal{U}(x) \in G.$$

On the other hand, since $k \in [v_G, w_G]$, we have

$$\nabla(k) = G.$$

Therefore,

$$\mathcal{U}(x) \in G \iff k(\mathcal{U}(x)) = S.$$

Using the adjunction $\mathcal{U} \dashv \mathcal{U}_*$ and the fact that $\mathcal{U}(1) = S$, we obtain

$$k(\mathcal{U}(x)) = S \iff \mathcal{U}_*(k(\mathcal{U}(x))) = 1.$$

Hence,

$$x \in F \iff (\mathcal{U}_* \circ k \circ \mathcal{U})(x) = 1 \iff x \in \nabla(\mathcal{U}_* \circ k \circ \mathcal{U}).$$

Thus,

$$\nabla(\mathcal{U}_* \circ k \circ \mathcal{U}) = F,$$

and consequently

$$\mathcal{U}_* \circ k \circ \mathcal{U} \in [v_F, w_F].$$

□

Proposition 4.2 defines a map

$$\Phi: [v_G, w_G] \longrightarrow [v_F, w_F], \quad \Phi(k) = \mathcal{U}_* \circ k \circ \mathcal{U}.$$

In particular, for every $k \in [v_G, w_G]$, we have

$$(2) \quad v_F \leq \mathcal{U}_* \circ k \circ \mathcal{U} \leq w_F.$$

Proposition 4.3. *Let (F, Q, G) be an HM triple. Then*

$$w_F = \mathcal{U}_* \circ w_G \circ \mathcal{U}.$$

Proof. By Proposition 4.2, since $w_G \in [v_G, w_G]$, we have

$$\mathcal{U}_* \circ w_G \circ \mathcal{U} \leq w_F.$$

Thus, it remains to prove the reverse inequality.

By adjunction, it is enough to show that

$$\mathcal{U} \circ w_F \leq w_G \circ \mathcal{U}.$$

Let $x \in A$. Since $w_G = [M']$, we have

$$w_G(\mathcal{U}(x)) = [M'](\mathcal{U}(x)) = (M' \cup \mathcal{U}(x))^\circ.$$

Hence, we need to prove that

$$\mathcal{U}(w_F(x)) \subseteq (M' \cup \mathcal{U}(x))^\circ.$$

Since $\mathcal{U}(w_F(x))$ is open, it is enough to show that

$$\mathcal{U}(w_F(x)) \subseteq M' \cup \mathcal{U}(x).$$

Suppose, by contradiction, that there exists

$$y \in \mathcal{U}(w_F(x))$$

such that

$$y \notin M' \quad \text{and} \quad y \notin \mathcal{U}(x).$$

Then $y \in M$ and $x \leq y$. On the other hand,

$$y \in \mathcal{U}(w_F(x))$$

implies

$$w_F(x) \not\leq y.$$

However,

$$w_F(x) = \bigwedge \{m \in M \mid x \leq m\},$$

and since $y \in M$ with $x \leq y$, it follows that

$$w_F(x) \leq y,$$

a contradiction.

Therefore,

$$\mathcal{U}(w_F(x)) \subseteq (M' \cup \mathcal{U}(x))^\circ,$$

and hence

$$\mathcal{U} \circ w_F \leq w_G \circ \mathcal{U}.$$

By adjunction,

$$w_F \leq \mathcal{U}_* \circ w_G \circ \mathcal{U}.$$

Combining both inequalities, we conclude that

$$w_F = \mathcal{U}_* \circ w_G \circ \mathcal{U}.$$

□

4.2. Stacked and strongly stacked frames.

Definition 4.4. Let A be a frame. We say that A is *stacked* if, for every HM pair (F, Q) ,

$$(U_* \circ [Q'] \circ U)(0) = v_F(0).$$

Definition 4.5. Let A be a frame. We say that A is *strongly stacked* if, for every HM pair (F, Q) ,

$$U_* \circ [Q'] \circ U = v_F;$$

Clearly, every strongly stacked frame is stacked.

4.3. KC frames.

Definition 4.6. Let A be a frame. We say that A is:

- (1) *KC* if every compact quotient of A is closed;
- (2) *Hausdorff closed compact*, or *KCH* for short, if every compact quotient of A is both closed and Hausdorff.

Equivalently, A is KC if, for every $j \in NA$ such that

$$\nabla(j) \in A^\wedge,$$

there exists $a \in A$ such that

$$j = u_a.$$

Similarly, A is KCH if, in addition, every such compact quotient A_j satisfies property (H) .

Proposition 4.7. *Every tidy strongly stacked frame is KC.*

Proof. Let $F \in A^\wedge$. Since A is tidy, it is T_1 . Hence, as A is strongly stacked, we have

$$v_F = w_F.$$

Moreover, of tidiness,

$$v_F = u_d, \quad d = v_F(0).$$

Therefore,

$$[v_F, w_F] = \{u_d\}.$$

Since every nucleus inducing a compact quotient of A belongs to an admissibility interval $[v_F, w_F]$ for some $F \in A^\wedge$, every such nucleus is closed. Hence every compact quotient of A is closed, and therefore A is KC. □

5. COMPARISON WITH CLASSICAL SEPARATION AXIOMS

Recall that if a topological space S is T_1 and strongly stacked, then

$$w_G = v_G.$$

where $G \in \mathcal{OS}^\wedge$. Moreover, every T_1 stacked space is sober. This naturally raises the question of whether analogous results hold in the point-free setting. In this section, we investigate this question.

Proposition 5.1. *Let A be a frame. If*

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$, then every point of A is maximal in $\text{pt } A$.

Proof. Let $p, q \in \text{pt } A$ be such that $p \leq q$, and consider the open filter

$$F_q = \{x \in A \mid x \not\leq q\}.$$

Since $A \setminus F_q = \downarrow q$, it follows directly from the definition of w_{F_q} that

$$(3) \quad w_{F_q} = w_q.$$

In particular,

$$w_{F_q}(0) = q.$$

By hypothesis,

$$q = w_{F_q}(0) = v_{F_q}(0) = \bigvee_{x \in F_q} \neg x.$$

For every $x \in F_q$, we have $x \not\leq q$. Since $p \leq q$, it follows that $x \not\leq p$. Moreover,

$$x \wedge \neg x = 0 \leq p.$$

Since p is a point, it is meet-irreducible, and therefore

$$\neg x \leq p.$$

Thus,

$$q = v_{F_q}(0) = \bigvee_{x \in F_q} \neg x \leq p.$$

Since $p \leq q$, we conclude that $p = q$. Hence, every point of A is maximal in $\text{pt } A$. $\square$

For every HM triple (F, Q, G) , we have

$$\mathcal{U} \circ v_F \leq v_G \circ \mathcal{U}.$$

Suppose now that A is strongly stacked. Then

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Since

$$v_G \leq [Q'],$$

we obtain

$$\mathcal{U}_* \circ v_G \circ \mathcal{U} \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} = v_F.$$

Together with the previous inequality, this yields

$$v_F = \mathcal{U}_* \circ v_G \circ \mathcal{U},$$

or equivalently,

$$\mathcal{U} \circ v_F = v_G \circ \mathcal{U}.$$

Thus, if A is strongly stacked, the corresponding diagram is commutative.

Proposition 5.2. *Let A be a tidy frame. If*

$$\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U}$$

for every HM triple (F, Q, ∇) , then A is strongly stacked.

Proof. Let $S = \text{pt } A$ and let (F, Q, ∇) be an HM triple.

Since A is tidy, $\mathcal{O}S$ is tidy. Hence S is stacked and packed. In particular, every compact saturated subset $Q \subseteq S$ is closed, so

$$Q' \in \mathcal{O}S.$$

Since S is stacked,

$$Q = \partial_\nabla^\infty(S),$$

and therefore, by Corollary 7.2,

$$v_\nabla(\emptyset) = Q'.$$

Because $\mathcal{O}S$ is tidy, it follows that

$$v_\nabla = u_{Q'} = [Q'].$$

By hypothesis,

$$\mathcal{U} \circ v_F = v_\nabla \circ \mathcal{U} = [Q'] \circ \mathcal{U}.$$

Applying the right adjoint $\mathcal{U}_*$, we obtain

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Therefore, A is strongly stacked. □

Consequently, we have the following result.

Corollary 5.3. *If A is a tidy frame, then $\text{pt } A$ is strongly stacked.*

Proof. Let $S = \text{pt } A$. Since A is tidy, $\mathcal{O}S$ is tidy, and therefore S is strongly stacked. □

Proposition 5.4. *Let A be a frame. The following statements hold:*

(1) *If A is T_1 and stacked, then*

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$.

(2) *If*

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$, then A is stacked.

(3) Suppose that

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$ and that

$$\{q \in \text{pt } A \mid a \leq q\} \neq \emptyset$$

for every $a \in A$. Then A is T_1 .

Proof.

(1) Let (F, Q) be an HM pair. Since A is T_1 , the space $S = \text{pt } A$ is T_1 . Hence $Q = M$. Since A is stacked,

$$\begin{aligned} v_F(0) &= (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) \\ &= (\mathcal{U}_* \circ [M'] \circ \mathcal{U})(0) \\ &= w_F(0). \end{aligned}$$

(2) Let (F, Q) be an HM pair. Recall that

$$(4) \quad v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq w_F.$$

Evaluating at 0, we obtain

$$v_F(0) \leq (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) \leq w_F(0).$$

By hypothesis, $v_F(0) = w_F(0)$, and hence

$$v_F(0) = (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0).$$

Therefore, A is stacked.

(3) Let $p \in \text{pt } A$ and $a \in A$ be such that

$$p \leq a.$$

By hypothesis, there exists $q \in \text{pt } A$ such that

$$a \leq q.$$

Hence,

$$p \leq a \leq q.$$

By Proposition 6.1, every point of A is maximal. Therefore, $p = q$, and consequently

$$p \leq a \leq p.$$

Thus, $a = p$, and hence A is T_1 . □

Thus, in the T_1 case, the stacked condition forces the lower and upper bounds of every admissibility interval to agree at 0. The strongly stacked case yields the corresponding equality of the nuclei themselves.

Corollary 5.5. *Let A be a frame. The following statements hold:*

(1) *If A is T_1 and strongly stacked, then*

$$v_F = w_F$$

for every $F \in A^\wedge$.

(2) If

$$v_F = w_F$$

for every $F \in A^\wedge$, then A is strongly stacked.

(3) Suppose that

$$v_F = w_F$$

for every $F \in A^\wedge$ and that

$$\{q \in \text{pt } A \mid a \leq q\} \neq \emptyset$$

for every $a \in A$. Then A is T_1 .

Proof.

(1) Let (F, Q) be an HM pair. Since A is strongly stacked,

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Moreover, since A is T_1 , we have $Q = M$. Hence, by Proposition 5.5,

$$v_F = \mathcal{U}_* \circ [M'] \circ \mathcal{U} = w_F.$$

(2) By (4),

$$v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq w_F.$$

Thus, if $v_F = w_F$, then

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U},$$

and therefore A is strongly stacked.

(3) Since $v_F = w_F$ implies

$$v_F(0) = w_F(0),$$

the result follows from Proposition 5.4 (3). □

While the strongly stacked condition yields the collapse of every admissibility interval, the stacked condition already guarantees this phenomenon for those associated with completely prime filters. This gives the following consequence.

Corollary 5.6. *Let A be a T_1 stacked frame. Then every admissibility interval associated with a completely prime filter collapses to a singleton. More precisely, for every $p \in \text{pt } A$,*

$$[v_{F_p}, w_{F_p}] = \{u_p\}.$$

Proof. Let $p \in \text{pt } A$. Since A is T_1 and stacked, Proposition 5.4 gives

$$v_{F_p}(0) = w_{F_p}(0) = p.$$

Moreover, as p is maximal,

$$w_{F_p} = w_p = u_p.$$

Recall that, for every nucleus j , one has

$$u_{j(0)} \leq j.$$

Indeed, for every $x \in A$,

$$j(0) \vee x \leq j(x).$$

Applying this to $j = v_{F_p}$ yields

$$u_p = u_{v_{F_p}(0)} \leq v_{F_p}.$$

On the other hand, by definition of the admissibility interval,

$$v_{F_p} \leq w_{F_p} = u_p.$$

Therefore,

$$[v_{F_p}, w_{F_p}] = \{u_p\}.$$

□

Notice that the additional hypothesis in Proposition 7.6 (3) is automatically satisfied whenever A is spatial. Indeed, if $a < 1$ and A is spatial, then there exists $p \in \text{pt } A$ such that

$$a \leq p.$$

Hence,

$$\mathcal{V}(a) = \{p \in \text{pt } A \mid a \leq p\} \neq \emptyset.$$

Thus, in the spatial setting, Proposition 7.6 reflects the topological fact that every T_1 stacked space is sober.

The point-free situation is, however, genuinely broader, since the T_1 and stacked conditions do not force spatiality. In fact, the following example shows that this remains true even if stacked is replaced by strongly stacked.

Example 5.7. Let B be an atomless complete Boolean algebra and consider

$$A = B \times 2,$$

where $2 = \{0 < 1\}$. Then A is a complete Boolean algebra. We show that A is T_1 and strongly stacked, but not spatial.

First, let

$$p = (1, 0).$$

We claim that

$$\text{pt } A = \{p\}.$$

Indeed, suppose that $(a, b), (c, d) \in A$ satisfy

$$(a, b) \wedge (c, d) \leq p.$$

Then

$$(a \wedge c, b \wedge d) \leq (1, 0),$$

so $b \wedge d = 0$. Since $b, d \in 2$, either $b = 0$ or $d = 0$, and hence

$$(a, b) \leq p \quad \text{or} \quad (c, d) \leq p.$$

Thus, $p \in \text{pt } A$.

Now, let $q = (x, y) \in \text{pt } A$. If $y = 0$, then

$$(1, 0) \wedge (x, 1) \leq (x, 0) = q.$$

Since $(x, 1) \not\leq q$, the complete meet-irreducibility of q implies

$$(1, 0) \leq q,$$

and therefore $x = 1$. Hence $q = p$.

If $y = 1$, then $x < 1$. For $a, b \in B$ such that $a \wedge b \leq x$, we have

$$(a, 1) \wedge (b, 1) \leq (x, 1) = q.$$

Since q is a point, either $a \leq x$ or $b \leq x$, showing that $x \in \text{pt } B$. This is impossible, since an atomless complete Boolean algebra has no points. Therefore,

$$\text{pt } A = \{p\}.$$

Moreover, if

$$p \leq (a, b),$$

then $a = 1$, and hence (a, b) is either p or 1 . Thus p is maximal and A is T_1 .

On the other hand, A is not spatial. Indeed, if $a, b \in B$ with $a \neq b$, then

$$(a, 0) \neq (b, 0),$$

while

$$\mathcal{U}(a, 0) = \emptyset = \mathcal{U}(b, 0).$$

Hence the spatial reflection

$$\mathcal{U}: A \longrightarrow \mathcal{O}(\text{pt } A)$$

is not injective.

It remains to prove that A is strongly stacked. Since B is an atomless complete Boolean algebra, 0 is its only compact element. Therefore,

$$K(A) = K(B) \times K(2) = \{(0, 0), (0, 1)\}.$$

Since every open filter of a complete Boolean algebra is principal and $\uparrow a$ is open if and only if a is compact, it follows that

$$A^\wedge = \{\uparrow(0, 0), \uparrow(0, 1)\}.$$

Clearly,

$$\uparrow(0, 0) = A,$$

whereas

$$\uparrow(0, 1) = B \times \{1\} = F_p.$$

Hence,

$$A^\wedge = \{A, F_p\}.$$

For F_p , since A is T_1 , then

$$v_{F_p} = w_{F_p}.$$

For $F = A$, we have

$$v_F = \text{tp} = w_F.$$

Consequently,

$$v_F = w_F$$

for every $F \in A^\wedge$. By Corollary 5.5, A is strongly stacked.

Corollary 5.8. *There exist non-spatial T_1 strongly stacked frames. In particular, there exist non-spatial T_1 stacked frames.*

The previous example shows that the T_1 and stacked conditions do not imply spatiality. We now compare stackedness with the stronger separation property (H) . Recall that a frame A satisfies property (H) if

$$1 \neq a \not\leq b$$

implies that there exists $c \in A$ such that

$$c \not\leq a \quad \text{and} \quad \neg c \not\leq b.$$

As shown in [PŠ92], property (H) implies T_1 . Nevertheless, property (H) does not imply stackedness, as the following example shows.

Example 5.9. There exists a frame satisfying property (H) which is not stacked.

Let $I = [0, 1]$, endowed with its usual topology, and consider the frame

$$A = \mathcal{O}(I).$$

Since I is a compact Hausdorff space, A is a compact Hausdorff frame. Moreover, since I is T_1 , every point of A is maximal and

$$D(A) = \text{pt } A = \{I \setminus \{x\} : x \in I\},$$

where $D(A)$ denotes the set of maximal elements of A .

Following the construction of Paseka and Šmarda [PŠ92, Proposition 2.3], let

$$\tilde{A} = \{(u, v) \in A \times \mathfrak{B}(A) : u \leq v\},$$

where

$$\mathfrak{B}(A) = \{a^{**} : a \in A\}$$

is the Boolean algebra of regular elements of A . By [PŠ92, Propositions 2.3 and 2.4], $\tilde{A}$ is a compact frame. In particular, $\tilde{A}$ satisfies property (H) .

The maximal elements of $\tilde{A}$ are precisely the elements of the form

$$D(\tilde{A}) = \{(d, 1) \mid d \in D(A)\}.$$

Since $\tilde{A}$ satisfies (H) , it is T_1 , and hence its points are precisely its maximal elements. Therefore,

$$\text{pt } \tilde{A} = \{(I \setminus \{x\}, I) : x \in I\}.$$

Since $\tilde{A}$ is compact,

$$F = \uparrow 1 = \{1\}$$

is an open filter. Let (F, Q) be the corresponding HM pair. By the Hofmann–Mislove correspondence,

$$Q = \text{pt } \tilde{A} - F.$$

Since $1 \notin \text{pt } \tilde{A}$, it follows that

$$Q = \text{pt } \tilde{A}.$$

Now, since $F = \{1\}$,

$$f_F = v_1 = \text{id}_{\tilde{A}},$$

and therefore

$$v_F(0) = 0_{\tilde{A}}.$$

On the other hand,

$$\begin{aligned} (U_* \circ [Q'] \circ U)(0) &= \bigwedge Q \\ &= \bigwedge_{\text{pt } \tilde{A}} \\ &= (\emptyset, I) \neq (\emptyset, \emptyset) = v_F(0). \end{aligned}$$

Thus,

$$v_F(0) \neq (U_* \circ [Q'] \circ U)(0),$$

and consequently $\tilde{A}$ is not stacked.

Consequently, although property (H) implies T_1 , it does not imply stackedness. Indeed, the preceding example provides a compact Hausdorff frame which is not stacked.

Together with the strongly stacked non-spatial example above, this further illustrates the distinction between separation properties and stackedness in the point-free setting. On the one hand, there exist non-spatial T_1 strongly stacked frames; on the other hand, even the stronger Hausdorff condition (H) does not guarantee stackedness.

6. QUOTIENTS AND PERMANENCE PROPERTIES

Proposition 6.1. *If A is a neat frame, then A_j is a neat frame.*

Proof. Let A be α -neat and let $F \in A_j^\wedge$. By Proposition 2.1, $j_*F \in A^\wedge$. Let

$$\hat{f} = \bigvee_{y \in j_*F} v_y \quad \text{and} \quad f = \bigvee_{x \in F} v_x.$$

be the prenuclei associated with j_*F and F , respectively.

Since A is neat, there exists an ordinal α such that, for every $y \in j_*F$,

$$y \vee \hat{d} = 1, \quad \text{with } \hat{d} = \hat{f}^\alpha(0).$$

Let

$$d = f^\alpha(j(0)),$$

by Corollary 6.7,

$$j(\hat{d}) = j(\hat{f}^\alpha(0)) \leq f^\alpha(j(0)) = d.$$

Since j is inflationary, $\hat{d} \leq j(\hat{d})$, and hence

$$\hat{d} \leq d.$$

Now, let $x \in F$. Thus, $x \in A_j$, we have $j(x) = x$, and therefore $x \in j_*F$. Then,

$$x \vee \hat{d} = 1.$$

As $\hat{d} \leq d$, it follows that

$$1 = x \vee \hat{d} \leq x \vee d \leq 1.$$

Hence $x \vee d = 1$. Therefore, A_j is α -neat. □

The following theorem establishes that tidiness is preserved under binary coproducts.

Proposition 6.2. *Let A_1 and A_2 be α -neat frames. Then $A_1 \oplus A_2$ is α -neat.*

Proof. By construction of the coproduct $A_1 \oplus A_2$, consider the canonical frame morphism

$$\iota_i: A_i \rightarrow A_1 \oplus A_2, \quad i = 1, 2$$

together with their right adjoints $(\iota_i)_*: A_1 \oplus A_2 \rightarrow A_i$.

Then, we have the following diagram

$$\begin{array}{ccccccc} & & & \iota_i & & & \\ & & \swarrow & & \searrow & & \\ A_i & \xrightarrow{\alpha_i} & A_1 \times A_2 & \xrightarrow{\downarrow(-)} & \mathcal{L}(A_1 \times A_2) & \xrightarrow{\sim} & A_1 \oplus A_2 \\ & & \nwarrow & & \nearrow & & \\ & & & (\iota_i)_* & & & \end{array}$$

By Proposition 2.1,

$$(\iota_i)_* F \in A_i^\wedge.$$

Since A_i is α -neat, for every $x_i \in (\iota_i)_* F$ there exists $d_i \in A_i$ such that

$$x_i \vee d_i = 1,$$

where

$$d_i = f_i^\alpha(0) \quad \text{and} \quad f_i = \bigvee_{y \in (\iota_i)_* F} v_y.$$

By the definition of $(\iota_i)_* F$,

$$x_i \in (\iota_i)_* F \iff \iota_i(x_i) \in F.$$

Thus,

$$x_1 \oplus 1, 1 \oplus x_2 \in F.$$

Since F is a filter,

$$(x_1 \oplus 1) \wedge (1 \oplus x_2) = x_1 \oplus x_2 \in F.$$

Now let

$$f_F = \bigvee_{z \in F} v_z$$

be the prenucleus associated with F . We first show that, for $i = 1, 2$,

$$\iota_i \circ f_i \leq f_F \circ \iota_i.$$

Indeed, let $a \in A_i$ and $y \in (\iota_i)_*F$. Since $\iota_i(y) \in F$,

$$\begin{aligned}\iota_i(v_y(a)) &= \iota_i(y \succ a) \\ &\leq (\iota_i(y) \succ \iota_i(a)) \\ &= v_{\iota_i(y)}(\iota_i(a)) \\ &\leq f_F(\iota_i(a)).\end{aligned}$$

Therefore, since the join defining f_i is pointwise and ι_i preserves arbitrary joins,

$$\iota_i(f_i(a)) = \iota_i\left(\bigvee_{y \in (\iota_i)_*F} v_y(a)\right) = \bigvee_{y \in (\iota_i)_*F} \iota_i(v_y(a)) \leq f_F(\iota_i(a)).$$

Hence,

$$\iota_i \circ f_i \leq f_F \circ \iota_i.$$

We now prove by transfinite induction that

$$\iota_i \circ f_i^\beta \leq f_F^\beta \circ \iota_i$$

for every ordinal β .

If $\beta = 0$, the result is immediate. Suppose that it holds for β . Then

$$\begin{aligned}\iota_i \circ f_i^{\beta+1} &= \iota_i \circ f_i \circ f_i^\beta \\ &\leq f_F \circ \iota_i \circ f_i^\beta \\ &\leq f_F \circ f_F^\beta \circ \iota_i \\ &= f_F^{\beta+1} \circ \iota_i.\end{aligned}$$

Now let λ be a limit ordinal. For every $a \in A_i$,

$$\begin{aligned}\iota_i(f_i^\lambda(a)) &= \iota_i\left(\bigvee_{\beta < \lambda} f_i^\beta(a)\right) \\ &= \bigvee_{\beta < \lambda} \iota_i(f_i^\beta(a)) \\ &\leq \bigvee_{\beta < \lambda} f_F^\beta(\iota_i(a)) \\ &= f_F^\lambda(\iota_i(a)).\end{aligned}$$

Thus,

$$\iota_i \circ f_i^\beta \leq f_F^\beta \circ \iota_i$$

for every ordinal β .

Evaluating at $0 \in A_i$ for $\beta = \alpha$, we obtain

$$\iota_1(d_1) \leq f_F^\alpha(0 \oplus 1) \quad \text{and} \quad \iota_2(d_2) \leq f_F^\alpha(1 \oplus 0).$$

Therefore,

$$\begin{aligned} d_1 \oplus d_2 &= (d_1 \oplus 1) \wedge (1 \oplus d_2) \\ &\leq f_F^\alpha(0 \oplus 1) \wedge f_F^\alpha(1 \oplus 0) \\ &= f_F^\alpha(0 \oplus 1 \wedge 1 \oplus 0). \end{aligned}$$

Set $d = f_F^\alpha(0 \oplus 0)$, then $d_1 \oplus d_2 \leq d$. Finally,

$$\begin{aligned} 1 \oplus 1 &= (x_1 \vee d_1) \oplus (x_2 \vee d_2) \\ &= (x_1 \oplus x_2) \vee (d_1 \oplus d_2) \\ &\leq (x_1 \oplus x_2) \vee d \\ &\leq 1 \oplus 1. \end{aligned}$$

Therefore,

$$(x_q \oplus x_2) \vee d = 1 \oplus 1$$

and $A_1 \oplus A_2$ is α -tidy. $\square$

Corollary 6.3. *Let A_1 and A_2 be α -neat and β -neat, respectively. Then $A_1 \oplus A_2$ is γ -neat, where*

$$\gamma = \max\{\alpha, \beta\}.$$

Proof. Let $\gamma = \max\{\alpha, \beta\}$. Then A_1 and A_2 are both γ -neat. Hence, by Proposition 6.2, $A_1 \oplus A_2$ is γ -neat. $\square$

We extend the previous result to arbitrary coproducts.

Proposition 6.4. *Tidiness is preserved under arbitrary coproducts.*

Proof. Let $\{A_i\}_{i \in I}$ be a family of neat frames, and for each $i \in I$, let α_i be a tidiness degree of A_i . Consider

$$A = \bigoplus_{i \in I} A_i$$

and let

$$\iota_i: A_i \rightarrow A$$

denote the canonical coproduct morphisms. Since ι_i is a frame morphism, it has a right adjoint $(\iota_i)_*$.

Let $F \in A^\wedge$. By Proposition 2.1,

$$(\iota_i)_* F \in A_i^\wedge$$

for every $i \in I$. Moreover,

$$x_i \in (\iota_i)_* F \iff \iota_i(x_i) \in F.$$

Set $\alpha = \sup\{\alpha_i \mid i \in I\}$. Then every A_i is α -neat. Hence, for every $x_i \in (\iota_i)_* F$,

$$x_i \vee d_i = 1,$$

where

$$d_i = f_i^\alpha(0) \quad \text{and} \quad f_i = \bigvee_{y \in (\iota_i)_* F} v_y.$$

Let

$$f_F = \bigvee_x \cdot \in Fv_x$$

be the prenucleus associated with F . As in the proof of Proposition 6.2, for every $i \in I$ we have

$$\iota_i \circ f_i \leq f_F \circ \iota_i,$$

and, by transfinite induction,

$$\iota_i \circ f_i^\beta \leq f_F^\beta \circ \iota_i$$

for every ordinal β .

Evaluating at $0 \in A_i$ for $\beta = \alpha$, we obtain

$$\iota_i(d_i) = \iota_i(f_i^\alpha(0)) \leq f_F^\alpha(\iota_i(0)) = f_F^\alpha(\langle 0 \rangle).$$

Set

$$d = f_F^\alpha(\langle 0 \rangle).$$

Therefore,

$$\iota_i(d_i) \leq d$$

for every $i \in I$. Taking arbitrary joins yields

$$\langle d_i \rangle \bigvee_{i \in I} \iota_i(d_i) \leq d.$$

Now let

$$\langle x_i \rangle \in F.$$

Since $x_i \vee d_i = 1$ for every $i \in I$, we have

$$\langle x_i \rangle \vee \langle d_i \rangle = \langle x_i \vee d_i \rangle = \langle 1 \rangle.$$

As $\langle d_i \rangle \leq d$,

$$\langle 1 \rangle = \langle x_i \rangle \vee \langle d_i \rangle \leq \langle x_i \rangle \vee d \leq \langle 1 \rangle.$$

Thus,

$$\langle x_i \rangle \vee d = \langle 1 \rangle.$$

Therefore, $A = \bigoplus_{i \in I} A_i$ is α -neat, and hence neat. $\square$

Corollary 6.5. *The subcategory of neat frames is a coreflective subcategory of $\mathbf{Frm}$.*

Simmons shows in [Sim04, Lemma 8.9 and Corollary 8.10] that the following diagram

$$(5) \quad \begin{array}{ccc} A & \xrightarrow{f^\infty} & A \\ U_A \downarrow & & \downarrow U_A \\ \mathcal{O}S & \xrightarrow{F^\infty} & \mathcal{O}S \end{array}$$

commutes laxly, that is, $U_A \circ f^\infty \leq F^\infty \circ U_A$. In this diagram f^∞ and F^∞ represent the fitted nuclei associated with the filters $F \in A^\wedge$ and $\nabla \in \mathcal{O}S^\wedge$, respectively, that is,

$$f^\infty = v_F \quad \text{and} \quad F^\infty = v_\nabla.$$

Here we prove something more general, since we consider $j \in NA$, $F \in A_j^\wedge$ and $j_*F \in A^\wedge$ to produce the diagram

$$\begin{array}{ccc} A & \xrightarrow{f^\infty} & A \\ j_* \downarrow & & \downarrow j_* \\ A_j & \xrightarrow{\hat{f}^\infty} & A_j \end{array}$$

where $\hat{f}^\infty$ and f^∞ are the fitted nuclei associated with the filters j_*F and F , respectively.

The following lemma is part of the analysis developed in this work and constitutes a key tool in the construction of the commutative diagrams that follow.

Lemma 6.6. *For $j = j_*$, f and $\hat{f}$ as before, it holds that $j \circ \hat{f} \leq f \circ j$.*

Proof. Recall that, by (1)

$$\hat{f} = \bigvee \{v_y \mid y \in j_*F\} \quad \text{y} \quad f = \bigvee \{v_{j(y)} \mid j(y) \in F\}.$$

Let $a \in A$ be. For every $y \in j_*F$, we have

$$j(v_y(a)) = j(y \succ a) \leq (j(y) \succ j(a)) = v_{j(y)}(j(a)) \leq f(j(a)),$$

where the last inequality follows from $j(y) \in F$. Therefore, since j preserves arbitrary joins,

$$j(\hat{f}(a)) = j\left(\bigvee \{v_y(a) \mid y \in j_*F\}\right) = \bigvee \{j(v_y(a)) \mid y \in j_*F\} \leq f(j(a)).$$

Hence, $j \circ \hat{f} \leq f \circ j$. □

We now prove the above for an arbitrary ordinal.

Corollary 6.7. *For j , f and $\hat{f}$ as before, it holds that $j \circ \hat{f}^\alpha \leq f^\alpha \circ j$ where $\alpha \in \text{Ord}$.*

Proof. We proceed by transfinite induction on α . If $\alpha = 0$, the result is immediate.

For the induction step, suppose that the inequality holds for α . Then

$$j \circ \hat{f}^{\alpha+1} = j \circ \hat{f} \circ \hat{f}^\alpha \leq f \circ j \circ \hat{f}^\alpha \leq f \circ f^\alpha \circ j = f^{\alpha+1} \circ j,$$

where the first inequality follows from Lemma 6.6 and the second from the induction hypothesis.

Finally, let λ be a limit ordinal. For every $a \in A$,

$$j(\hat{f}(a)) = j\left(\bigvee_{\alpha < \lambda} \hat{f}^\alpha\right) = \bigvee_{\alpha < \lambda} j(\hat{f}^\alpha(a)) \leq \bigvee_{\alpha < \lambda} f^\alpha(j(a)) = f^\lambda(j(a)).$$

where the inequality follows from the induction hypothesis. Hence

$$j \circ \hat{f}^\lambda \leq f^\lambda \circ j.$$

This completes the proof. $\square$

Recalling that $v_F = f^\infty$ and $v_{j_*F} = \hat{f}^\infty$, Corollary 6.7 yields

$$j \circ v_{j_*F} \leq v_F \circ j$$

We now compare the compact quotients induced by F and j_*F , and construct a frame morphism between them making the corresponding diagram commute.

If we consider j, v_F and v_{j_*F} we have the following diagram

$$(6) \quad \begin{array}{ccc} A & \xrightleftharpoons[(v_{j_*F})_*]{(v_{j_*F})^*} & A_{j_*F} \\ j \downarrow & \searrow H & \downarrow h \\ A_j & \xrightleftharpoons[(v_F)^*]{(v_F)_*} & A_F \end{array}$$

where A_F and A_{j_*F} are the compact quotients produced by v_F and v_{j_*F} , respectively.

Define

$$H: A \rightarrow A_F, \quad H = v_F \circ j.$$

Since v_F is idempotent, $H(A) \subseteq A_F$. Hence, by restricting H to A_{j_*F} , we obtain a map

$$h: A_{j_*F} \rightarrow A_F, \quad h = H|_{A_{j_*F}}.$$

As finite meets in quotient frames are inherited from the ambient frame, h preserves finite meets. It remains to verify that h preserves arbitrary joins.

Lemma 6.8. *The map $h: A_{j_*F} \rightarrow A_F$ preserves arbitrary joins.*

Proof. Let $X \subseteq A_{j_*F}$. Since joins in A_{j_*F} are given by

$$\bigvee_{v_{j_*F}} X = v_{j_*F}(\bigvee X)$$

we have

$$\begin{aligned} h\left(\bigvee_{v_{j_*F}} X\right) &= v_F\left(j\left(v_{j_*F}\left(\bigvee X\right)\right)\right) \\ &\leq v_F\left(v_F\left(j\left(\bigvee X\right)\right)\right) \\ &= v_F\left(j\left(\bigvee X\right)\right), \end{aligned}$$

where the inequality follows from Corollary 6.7 and the equality from the idempotency of v_F .

Since H is a frame morphism,

$$v_F \left(j \left(\bigvee X \right) \right) = \bigvee_{v_F} \{H(x) \mid x \in X\} = \bigvee_{v_F} \{h(x) \mid x \in X\}.$$

Therefore,

$$h \left(\bigvee_{v_{j_* F}} X \right) \leq \bigvee_{v_F} h[X].$$

Conversely, for every $x \in X$,

$$x \leq \bigvee_{v_{j_* F}} X,$$

and hence, by monotonicity of h ,

$$h(x) \leq h \left(\bigvee_{v_{j_* F}} X \right).$$

Since this holds for every $x \in X$, we have

$$\bigvee_{v_F} h[X] \leq h \left(\bigvee_{v_{j_* F}} X \right).$$

Thus,

$$h \left(\bigvee_{v_{j_* F}} X \right) = \bigvee_{v_F} h[X].$$

Therefore h preserves arbitrary joins. $\square$

The previous construction yields the following commutative diagram induced by the compact quotients associated with the fitted nuclei.

Proposition 6.9. *The diagram*

$$(7) \quad \begin{array}{ccc} A & \xrightarrow{v_{j_* F}} & A_{j_* F} \\ j \downarrow & & \downarrow h \\ A_j & \xrightarrow{v_F} & A_F \end{array}$$

is commutative.

Proof. By Lemma 6.8, h is a frame morphism. The commutativity of the diagram follows from the definition of h . $\square$

We now study the behavior of strongly stacked frames under quotients.

Recall that, for a frame morphism $f: A \rightarrow B$, $j \in NA$, and $k \in NB$, we have

$$N(f)(j) = \bigvee \{u_{f(j(a))} \wedge v_{f(a)} \mid a \in A\} \in NB$$

and

$$N(f)_*(k) = f_* \circ k \circ f \in NA,$$

where f_* denotes the right adjoint of f .

Proposition 6.10. *Let A be a strongly stacked frame and $j \in NA$. Then A_j is strongly stacked.*

Proof. Let $F \in A_j^\wedge$, and set

$$S = \text{pt } A, \quad S_j = \text{pt } A_j.$$

Consider the open filter

$$G = j_*(F) = \{x \in A \mid j(x) \in F\},$$

and let

$$Q = S_j \setminus F, \quad R = S \setminus G.$$

Denote by

$$\mathcal{U}: A \longrightarrow \mathcal{O}S \quad \text{and} \quad \mathcal{U}_j: A_j \longrightarrow \mathcal{O}S_j$$

the corresponding spatial reflections.

Set

$$l = N(\mathcal{U}_j)_*([Q']) = (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j$$

and

$$m = N(j^*)(v_G).$$

By Proposition 4.2,

$$v_F \leq l.$$

Thus, it is enough to prove

$$l \leq m \leq v_F.$$

We first prove that $m \leq v_F$. Since $G = j_*(F)$, the fitted nuclei associated with G and F satisfy

$$(8) \quad j^* \circ v_G \leq v_F \circ j^*.$$

Moreover,

$$m = N(j^*)(v_G) = \bigvee_{x \in A} (u_{j^*(v_G(x))} \wedge v_{j^*(x)}).$$

By (8), for every $x \in A$,

$$j^*(v_G(x)) \leq v_F(j^*(x)),$$

and hence

$$u_{j^*(v_G(x))} \leq u_{v_F(j^*(x))}.$$

Therefore,

$$u_{j^*(v_G(x))} \wedge v_{j^*(x)} \leq u_{v_F(j^*(x))} \wedge v_{j^*(x)} \leq v_F.$$

Taking joins over $x \in A$, we obtain

$$(9) \quad m \leq v_F.$$

We now prove that $l \leq m$. Since A is strongly stacked,

$$v_G = N(\mathcal{U})_*([R']) = \mathcal{U}_* \circ [R'] \circ \mathcal{U}.$$

Hence,

$$m = N(j^*)(N(\mathcal{U})_*([R'])).$$

To compare the u -components in the decompositions of l and m , consider the auxiliary nucleus

$$n = N(\mathcal{U}_j \circ j^*)_*([Q']).$$

We first observe that, for every $x \in A_j$,

$$j^*(n(x)) = l(x).$$

Indeed,

$$\begin{aligned} j^*(n(x)) &= j^*(j_* \circ (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j \circ j^*)(x) \\ &= j^*j_* \left(\bigvee \{a \in A_j \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(j^*(x))\} \right) \\ &= \bigvee \{a \in A_j \mid \mathcal{U}_j(a) \subseteq Q' \cup \mathcal{U}_j(x)\} \\ &= l(x). \end{aligned}$$

Now, under the usual identification of S_j with the corresponding subspace of S , we have

$$Q = R \cap S_j.$$

Using the decomposition of a nucleus in terms of its u - and v -components, we have

$$l = \bigvee_{x \in A_j} (u_{l(x)} \wedge v_x) = \bigvee_{x \in A_j} (u_{j^*(n(x))} \wedge v_x).$$

On the other hand, since

$$m = N(j^*)(N(\mathcal{U})_*([R'])),$$

we have

$$m = \bigvee_{a \in A} (u_{j^*(N(\mathcal{U})_*([R'])(a))} \wedge v_{j^*(a)}).$$

For $x \in A_j$, the explicit description of the right adjoint gives

$$\begin{aligned} j^*(N(\mathcal{U})_*([R'])(x)) &= j^* \left(\bigvee \{a \in A \mid \mathcal{U}(a) \subseteq R' \cup \mathcal{U}(x)\} \right) \\ &= \bigvee \{j^*(a) \in A_j \mid \mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)\}. \end{aligned}$$

Moreover, since $x \in A_j$,

$$\mathcal{U}(j^*(a)) \subseteq Q' \cup \mathcal{U}(x)$$

is equivalent, after restriction to S_j , to

$$\mathcal{U}_j(j^*(a)) \subseteq Q' \cup \mathcal{U}_j(x).$$

Consequently,

$$j^*(n(x)) \leq j^*(N(\mathcal{U})_*([R'])(x)).$$

Therefore,

$$u_{j^*(n(x))} \leq u_{j^*(N(\mathcal{U})_*([R'])(x))}$$

for every $x \in A_j$, and hence

$$u_{l(x)} \wedge v_x = u_{j^*(n(x))} \wedge v_x \leq u_{j^*(N(\mathcal{U})_*([R'])(x))} \wedge v_x \leq m.$$

Taking joins over $x \in A_j$, we obtain

$$l \leq m.$$

Combining this inequality with (9) and $v_F \leq l$, we obtain

$$v_F \leq l \leq m \leq v_F.$$

Hence,

$$v_F = l = (\mathcal{U}_j)_* \circ [Q'] \circ \mathcal{U}_j.$$

Therefore, A_j is strongly stacked. $\square$

Since every strongly stacked frame is stacked, we obtain the following result.

Corollary 6.11. *Let A be a strongly stacked frame and $j \in NA$. Then A_j is stacked.*

Proof. By Proposition 6.10, A_j is strongly stacked, and hence stacked. $\square$

We first observe that the KC property is inherited by quotients.

Proposition 6.12. *Let A be a KC frame and $j \in NA$. Then A_j is KC.*

Proof. Let $k \in N(A_j)$ be such that $(A_j)_k$ is compact. Then

$$\nabla(k) \in A_j^\wedge.$$

By Proposition 2.1,

$$j_*(\nabla(k)) \in A^\wedge.$$

Consider the nucleus

$$l = j_* \circ k \circ j^* \in NA.$$

Since

$$\nabla(l) = j_*(\nabla(k)),$$

the quotient A_l is compact. Since A is KC, there exists $a \in A$ such that

$$l = u_a.$$

We claim that

$$k = u_{j^*(a)}.$$

Indeed, from $l = u_a$ we have

$$a \vee x = l(x) = j_*(k(j^*(x)))$$

for every $x \in A$. Applying j^* to both sides yields

$$j^*(a) \vee j^*(x) = k(j^*(x)),$$

since $j^* \circ j_* = \text{id}_{A_j}$.

Now let $x \in A_j$. Since $j^*: A \rightarrow A_j$ is surjective, we have $j^*(x) = x$, and therefore

$$\begin{aligned} u_{j^*(a)}(x) &= j^*(a) \vee x \\ &= j^*(a) \vee j^*(x) \\ &= k(j^*(x)) \\ &= k(x). \end{aligned}$$

Hence

$$k = u_{j^*(a)}.$$

Thus $(A_j)_k$ is closed. Therefore, A_j is KC. □

Proposition 6.13. *Every KC frame is T_1 .*

Proof. Let $p \in \text{pt } A$. The filter

$$F_p = \nabla(w_p) = \{x \in A \mid x \not\leq p\}$$

is completely prime and, in particular, $F_p \in A^\wedge$. Hence A_{w_p} is a compact quotient of A . Since A is KC, this quotient is closed. Therefore,

$$w_p = u_a$$

for some $a \in A$. Evaluating at 0, we obtain

$$a = u_a(0) = w_p(0) = p,$$

and hence

$$w_p = u_p.$$

Now let $a \in A$ be such that

$$p < a \leq 1.$$

Since $a \not\leq p$, we have

$$w_p(a) = 1.$$

On the other hand,

$$u_p(a) = p \vee a = a.$$

Since $w_p = u_p$, it follows that

$$a = 1.$$

Thus p is maximal. Since every point of A is maximal, A is T_1 . □

7. THE SPATIAL CASE

Lemma 7.1. *Let $\alpha \in \text{Ord}$ and $X \in CS$. Then*

$$(10) \quad \partial_F^\alpha(X) = (f_F^\alpha(X'))'.$$

Proof. We proceed by transfinite induction on α .

For $\alpha = 0$, both ∂_F^0 and f_F^0 are the identity. Hence,

$$\partial_F^0(X) = X = (X')' = (f_F^0(X'))'.$$

Suppose that the result holds for some ordinal α . Then

$$\begin{aligned} \partial_F^{\alpha+1}(X) &= \partial_F(\partial_F^\alpha(X)) \\ &= (f_F((\partial_F^\alpha(X))'))' \\ &= (f_F(f_F^\alpha(X')))' \\ &= (f_F^{\alpha+1}(X'))'. \end{aligned}$$

Finally, let λ be a limit ordinal and suppose that the result holds for every $\alpha < \lambda$. By definition and De Morgan's law,

$$\begin{aligned}
 (\partial_F^\lambda(X))' &= \left(\bigcap_{\alpha < \lambda} \partial_F^\alpha(X) \right)' \\
 &= \bigcup_{\alpha < \lambda} (\partial_F^\alpha(X))' \\
 &= \bigcup_{\alpha < \lambda} f_F^\alpha(X') \\
 &= f_F^\lambda(X'),
 \end{aligned}$$

where the third equality follows from the induction hypothesis. Therefore,

$$\partial_F^\lambda(X) = \left(f_F^\lambda(X') \right)'.$$

This completes the proof. $\square$

The following is an immediate consequence of Lemma 7.1.

Corollary 7.2. *Let $U \in \mathcal{OS}$. Then*

$$(11) \quad (\partial_F^\infty(U'))' = v_F(U).$$

A particular instance of diagram (7) is obtained by taking $j^* = \mathcal{U}_A$, where $S = \text{pt } A$. Thus, we obtain the following commutative diagram:

$$\begin{array}{ccc}
 A & \xrightarrow{v_F} & A_F \\
 \mathcal{U}_A \downarrow & & \downarrow h \\
 \mathcal{OS} & \xrightarrow{v_\nabla} & \mathcal{OS}_\nabla
 \end{array}$$

Recall that $\text{pt } A_F = Q$. Hence, the spatial reflection of A_F is given by

$$\mathcal{U}_{A_F}: A_F \rightarrow \mathcal{O} \text{pt } A_F = \mathcal{O}Q.$$

Similarly, since $\text{pt}((\mathcal{OS})_\nabla) = Q$, we have the spatial reflection

$$\mathcal{U}_{\mathcal{OS}_\nabla}: \mathcal{OS}_\nabla \rightarrow \mathcal{O} \text{pt}(\mathcal{OS})_\nabla = \mathcal{O}Q.$$

Consequently, the two induced morphisms from A to $\mathcal{O}Q$ are both given by

$$x \mapsto \mathcal{U}_A(x) \cap Q.$$

Therefore, the preceding constructions fit into the following commutative diagram:

$$(12) \quad \begin{array}{ccccc}
 A & \xrightarrow{v_F} & A_F & & \\
 \mathcal{U}_A \downarrow & & \downarrow h & \searrow \mathcal{U}_{A_F} & \\
 & & & & \mathcal{O}Q \\
 \mathcal{OS} & \xrightarrow{v_\nabla} & \mathcal{OS}_\nabla & \nearrow \mathcal{U}_{\mathcal{OS}_\nabla} & \\
 & & & &
 \end{array}$$

Since the above diagram is determined by $F \in A^\wedge$ and the associated compact saturated subset $Q \in \mathcal{Q}S$, we refer to diagram (12) as the Q -square.

Theorem 7.3. *Under the hypotheses of the Q -square, if A satisfies property **(H)**, then*

$$\mathcal{O}Q \cong \uparrow Q'.$$

Equivalently, the frame of open sets of the point space of A_F is isomorphic to a compact closed quotient of a Hausdorff space.

Proof. Suppose that A satisfies **(H)**. Since **(H)** is conservative, $S = \text{pt } A$ is a T_2 space. Hence, for every $Q \in \mathcal{Q}S$, the set Q is closed, and therefore $Q' \in \mathcal{O}S$

Moreover, since every T_2 space is strongly stacked, we have

$$v_\nabla = [Q'].$$

As $Q' \in \mathcal{O}S$, it follows that

$$[Q'] = u_{Q'}$$

Consequently,

$$(\mathcal{O}S)_{v_\nabla} = (\mathcal{O}S)_{u_{Q'}} \cong \uparrow Q'.$$

Now, $u_{Q'}$ is a complemented nucleus in $\mathcal{O}S$, by [PP12, Proposition VI.3.3], implies that $\mathcal{O}S_{v_\nabla}$ is spatial. Therefore,

$$(\mathcal{O}S)_{v_\nabla} \cong \mathcal{O} \text{pt}((\mathcal{O}S)_{v_\nabla}).$$

Since

$$\text{pt}((\mathcal{O}S)_{v_\nabla}) = Q,$$

we obtain

$$\mathcal{O}Q \cong (\mathcal{O}S)_{v_\nabla} \cong \uparrow Q'.$$

□

Lemma 7.4. *Let $A = \mathcal{O}S$ be spatial and let $E \subseteq S$. Then*

$$\mathcal{U}_* \circ [E] \circ \mathcal{U} = [E].$$

Proof. Let $V \in \mathcal{O}S$. Since

$$\mathcal{U}: \mathcal{O}S \longrightarrow \mathcal{O} \text{pt}(\mathcal{O}S)$$

is an isomorphism, there exist $x, y \in \mathcal{O}S$ such that

$$V = \mathcal{U}(y)$$

and

$$(E \cup V)^\circ = \mathcal{U}(x).$$

Then

$$\begin{aligned} (\mathcal{U}_* \circ [E] \circ \mathcal{U})(y) &= \mathcal{U}_*([E](\mathcal{U}(y))) \\ &= \mathcal{U}_*((E \cup V)^\circ) \\ &= \mathcal{U}_*(\mathcal{U}(x)) \\ &= x. \end{aligned}$$

On the other hand,

$$\mathcal{U}([E](y)) = (E \cup \mathcal{U}(y))^\circ = (E \cup V)^\circ = \mathcal{U}(x).$$

Since $\mathcal{U}$ is injective, it follows that

$$[E](y) = x.$$

Therefore,

$$(\mathcal{U}_* \circ [E] \circ \mathcal{U})(y) = [E](y)$$

for every $y \in \mathcal{O}S$. Hence,

$$\mathcal{U}_* \circ [E] \circ \mathcal{U} = [E].$$

□

We now show that both notions are conservative; namely, for every topological space S ,

$$S \text{ is stacked} \iff \mathcal{O}S \text{ is stacked},$$

and

$$S \text{ is strongly stacked} \iff \mathcal{O}S \text{ is strongly stacked}.$$

Proposition 7.5. *The notions of strongly stacked and stacked are conservative.*

Proof. Let $S \in \mathbf{Top}$ and set $A = \mathcal{O}S$.

First, suppose that S is strongly stacked. Then, for every HM pair (F, Q) ,

$$v_F = [Q'].$$

By Lemma 5.3,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = [Q'].$$

Therefore,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = v_F,$$

and hence $\mathcal{O}S$ is strongly stacked.

Now, suppose that S is stacked. Then, for every HM pair (F, Q) ,

$$Q = \partial_F^\infty(S).$$

Hence,

$$Q' = (\partial_F^\infty(S))'.$$

By Corollary 5.2,

$$v_F(\emptyset) = (\partial_F^\infty(S))' = Q'.$$

In particular, Q' is open. By Lemma 5.3,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = [Q'].$$

Therefore,

$$\begin{aligned} (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(\emptyset) &= [Q'](\emptyset) \\ &= (Q')^\circ \\ &= Q' \\ &= v_F(\emptyset). \end{aligned}$$

Thus, $\mathcal{O}S$ is stacked.

Since all the implications used above are equivalences, the converse statements follow by reversing the argument. $\square$

Recall that $A^\wedge$ is endowed with the topology generated by the basic open sets

$$D(x) = \{F \in A^\wedge \mid x \in F\}, \quad x \in A.$$

Moreover, let VA denote the Vietoris modification of A introduced in [Sim04]. The points of VA are of the form

$$(F, a), \quad F \in A^\wedge, \quad a \in [v_F(0), w_F(0)].$$

Consider the continuous surjection

$$\pi: \text{pt}(VA) \longrightarrow A^\wedge, \quad (F, a) \longmapsto F.$$

It induces the frame morphism

$$\mathcal{O}(\pi) = \pi^{-1}: \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\text{pt}(VA)).$$

Recall also that the subbasic open sets of $\text{pt}(VA)$ are determined by

$$(13) \quad (F, a) \in \Box x \iff x \in F, \quad (F, a) \in \Diamond x \iff x \not\leq a.$$

Particularly,

$$(14) \quad a \in [F](x) \iff x \in F \quad \text{and} \quad a \in \langle F \rangle(x) \iff x \not\leq a.$$

Proposition 7.6. *Let A be a frame, and consider the continuous surjection*

$$\pi: \text{pt}(VA) \longrightarrow A^\wedge, \quad (F, a) \longmapsto F.$$

Then the following statements hold:

- (1) *If A is T_1 and stacked, then*

$$\mathcal{O}(\pi) = \pi^{-1}: \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\text{pt}(VA))$$

is an isomorphism.

- (2) *If $\mathcal{O}(\pi)$ is an isomorphism, then A is stacked.*

- (3) *If $\mathcal{O}(\pi)$ is an isomorphism and*

$$\mathcal{V}(a) = \{p \in \text{pt } A \mid a \leq p\} \neq \emptyset$$

for every $a < 1$, then A is T_1 .

Proof.

- (1) Suppose that A is T_1 and stacked. By Proposition 5.4,

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$. Hence, each fiber of

$$\pi: \text{pt}(VA) \longrightarrow A^\wedge$$

contains a unique element. Therefore, π is bijective.

It remains to prove that

$$\pi^{-1}: A^\wedge \longrightarrow \text{pt}(VA)$$

is continuous. Recall that the subbasic open sets of $\text{pt}(VA)$ are given by

$$(F, a) \in [-](x) \iff x \in F$$

and

$$(F, a) \in \langle - \rangle(x) \iff x \not\leq a.$$

For the first family,

$$(F, a) \in [-](x) \iff x \in F \iff F \in D(x),$$

and therefore

$$\pi([-](x)) = D(x).$$

Now consider $\langle - \rangle(x)$. Since the fibers of π collapse, for every $(F, a) \in \text{pt}(VA)$ we have

$$a = v_F(0).$$

Thus,

$$\begin{aligned} (F, a) \notin \langle - \rangle(x) &\iff x \leq a \\ &\iff x \leq v_F(0) \\ &\iff v_F(x) = v_F(0) \\ &\iff x \notin F. \end{aligned}$$

Hence,

$$(F, a) \in \langle - \rangle(x) \iff x \in F \iff F \in D(x),$$

and therefore

$$\pi(\langle - \rangle(x)) = D(x).$$

Thus, the image under π of every subbasic open set of $\text{pt}(VA)$ is open in $A^\wedge$. Hence π^{-1} is continuous, and consequently π is a homeomorphism. Therefore,

$$\mathcal{O}(\pi) = \pi^{-1}: \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\text{pt}(VA))$$

is an isomorphism.

(2) Suppose that

$$\mathcal{O}(\pi) = \pi^{-1}: \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\text{pt}(VA))$$

is an isomorphism. Since π^{-1} is bijective, π is injective. For every $F \in A^\wedge$,

$$(F, v_F(0)), (F, w_F(0)) \in \text{pt}(VA)$$

and

$$\pi(F, v_F(0)) = F = \pi(F, w_F(0)).$$

Hence,

$$v_F(0) = w_F(0).$$

By Proposition 5.4, A is stacked.

(3) Suppose that $\mathcal{O}(\pi)$ is an isomorphism and that

$$\mathcal{V}(a) = \{q \in \text{pt } A \mid a \leq q\} \neq \emptyset$$

for every $a < 1$. By part (2),

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$. Therefore, by Proposition 5.1, every point of A is maximal.

Let $p \in \text{pt } A$ and $a \in A$ be such that

$$p \leq a.$$

By hypothesis, there exists $q \in \text{pt } A$ such that

$$a \leq q.$$

Thus,

$$p \leq a \leq q.$$

Since p and q are maximal, we have $p = q$, and therefore

$$p \leq a \leq p.$$

Hence $a = p$, and A is T_1 . □

8. RELATIONS AMONG COMPACT-QUOTIENT SEPARATION PROPERTIES

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Following the idea presented in [Sim06b] (or, alternatively, in [Sim04]), let us consider the linearly ordered frame

$$A = [0, 1].$$

In this case the following observations hold.

- (1) For all $a, b \in A$ the implication $(a \succ b)$ is given by

$$(a \succ b) = \begin{cases} 1, & \text{si } a \leq b, \\ b, & \text{si } a > b. \end{cases}$$

- (2) For $a, x \in A$, the distinguished nuclei u_a, v_a, w_a are given by

$$u_a(x) = \begin{cases} x & \text{si } a \leq x, \\ a & \text{si } x < a, \end{cases}, \quad v_a(x) = \begin{cases} 1 & \text{si } x \geq a, \\ a & \text{si } x < a, \end{cases}, \quad w_a(x) = \begin{cases} 1 & \text{si } x > a, \\ a & \text{si } x \leq a. \end{cases}$$

- (3) If $F \subseteq A$ is a filter, then F is always admissible and has the form

$$F = \nabla(w_a) = (a, 1] \quad \text{o} \quad F = \nabla(v_a) = [a, 1]$$

for some $a \in A$. In particular, $F \in A^\wedge$ if and only if $F = (a, 1]$.

- (4) If $F \in A^\wedge$, the adjusted nucleus associated v_F is calculated as

$$(15) \quad v_F(x) = l_m(x) = \begin{cases} 1 & \text{if } x > m, \\ x & \text{if } x \leq m. \end{cases}$$

where $l_m = v_m \wedge w_m$ and $m = \bigwedge F$.

- (5) For $S = \text{pt } A$ it holds that $S = [0, 1)$ and the open sets in $\mathcal{O}S$ are of the form

$$\mathcal{U}_A(x) = \{p \in S \mid p < x\} = [0, x).$$

- (6) The frame A is not T_1 , since no $p \in S$ is maximal. Consequently, A does not satisfy any stronger separation property in Frm .
 (7) For $F \in A^\wedge$ the compact saturated set associated $Q \in \mathcal{Q}S$ is of the form $Q = [0, m]$.
 (8) It holds that $[Q'] = v_F$; therefore, S is strongly stacked.
 (9) Since $[0, x) \in \mathcal{O}S$, then $[x, 1] \in \mathcal{C}S$ and $Q = [0, m]$. Therefore, the compact subsets are not closed; that is, S is not packed.
 (10) The frame A is not efficient.

The above resume some characteristics about the frame $[0, 1]$. Now, we will perform the analysis of the patch constructions of A and its point space S .

Remember that for $A \in \text{Frm}$ and $S \in \text{Top}$ we have

$$\text{Pbase} = \{u_a \wedge v_F \mid a \in A, F \in A^\wedge\} \quad \text{y} \quad \text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}$$

where Pbase generates the patch frame of A and pbase generates the topology of the patch space of S . By (15) we have that

$$\text{Pbase} = \{u_a \wedge l_m \mid a \in [0, 1], m \in [0, 1)\}.$$

For $x \in A$ we have

$$(u_a \wedge l_m)(x) = u_a(x) \wedge l_m(x).$$

Let us analyze the value of this map according to the relative position of x and m .

Case 1: If $x < m$, then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge x = x.$$

Case 2: If $x = m$, then

$$(u_a \wedge l_m)(x) = u_a(m) \wedge m = x.$$

Case 3: If $x > m$, then

$$(u_a \wedge l_m)(x) = u_a(x) \wedge 1 = u_a(x).$$

In the first two cases, the value of a does not affect the evaluation. Thus, the only non-trivial case occurs when $m < x$, where the value of $(u_a \wedge l_m)(x)$ depends on the relative position of a and x . We therefore distinguish the following two subcases.

Case 3.1: If $m < x < a$, then

$$(u_a \wedge l_m)(x) = a.$$

Case 3.2: If $m < a \leq x$, then

$$(u_a \wedge l_m)(x) = x.$$

From the preceding analysis, we conclude that the non-trivial generators of Pbase correspond precisely to the pairs (m, a) with $m < a$. Consequently, the frame PA is determined by the open intervals of the form (m, a) , where $m \in \text{pt } A$ and $a \in A$.

We now turn to the question of whether PA satisfies any separation property. To this end, we recall the relationship between the point-free patch construction and its spatial counterpart.

Let A be an arbitrary frame and let $S = \text{pt } A$ be its space of points. The two patch constructions are related by the following commutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & PA & \hookrightarrow & NA \\ \mathcal{U}_A \downarrow & & \downarrow P(\mathcal{U}_A) & & \downarrow N(\mathcal{U}_A) \\ \mathcal{O}S & \longrightarrow & POS & \hookrightarrow & NOS \\ & \searrow & \downarrow \pi & & \downarrow \sigma \\ & & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S \end{array}$$

where $\mathcal{O}^f S$ is the Skula topology (or front topology) of S . In [SS06b] is named *the patch assembly diagram* and more details about it can also be found in [Sex03].

In our case, $\mathcal{U}_A$ is an isomorphism. Moreover, under $\mathcal{U}_A$, the patch construction turns out to be functorial and, therefore, $PA \cong P\mathcal{O}S$. Similarly, $NA \cong N\mathcal{O}S$. To describe explicitly $P\mathcal{O}S$ and $N\mathcal{O}S$, we use the following result, which can be consulted in [ÁBMZ20].

Proposition 8.1. *Let S be a partially ordered set.*

- (1) *If S has no infinite antichains, then $N\mathcal{O}S$ is spatial.*
- (2) *If S is totally ordered, then $N\mathcal{O}S$ is spatial.*

Thus, by Proposition 8.1 we have that $N\mathcal{O}S$ is spatial and therefore $N\mathcal{O}S \cong \mathcal{O}^f S$. With all this information, the patch assembly diagram, in our particular case is as follows.

$$\begin{array}{ccccc} A & \longrightarrow & PA & \longrightarrow & NA \\ \cong \downarrow & & \downarrow \cong & & \downarrow \cong \\ \mathcal{O}S & \longrightarrow & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

Therefore, if we want to study the frame PA , we can study the patch topology $\mathcal{O}^p S$. Remember that

$$\text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}.$$

In our case,

$$\text{pbase} = \{[0, a) \cap [m, 1) \mid a \in [0, 1], m \in [0, 1)\} = \{(m, a)\}.$$

Since the topology generated by the intervals (m, a) coincides with the usual topology on $[0, 1)$, the space ${}^p S$ is Hausdorff. Consequently, $\mathcal{O}^p S \cong PA$ satisfies **(H)**, even when A is not T_1 nor tidy.

Lemma 8.2. *Let A be a frame, if the morphism $p\mathcal{U}: pA \rightarrow p\mathcal{O}S$ is an isomorphism, then A is strongly stacked.*

Proof. Consider any open filter F on A and the compact saturated determined by it, say Q , thus we have the nuclei v_F and $U_*[Q']U = Q[U]$, we need to observe the action of $p\mathcal{U}$ on these two nuclei, in order to do that remember that we have a commutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & pA & \xleftarrow{\iota} & NA \\ \mathcal{U}_A \downarrow & & \downarrow p(\mathcal{U}_A) & & \downarrow N(\mathcal{U}_A) \\ \mathcal{O}S & \longrightarrow & p\mathcal{O}S & \xleftarrow{\iota} & N\mathcal{O}S \\ & \searrow & \downarrow \pi & & \downarrow \sigma \\ & & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

since $N(\mathcal{U})\iota = p\mathcal{U}$, that is, $p\mathcal{U}$ is the restriction of $N(\mathcal{U})$ to pA , then

$$p\mathcal{U}(v_F) = N(\mathcal{U})(v_F) = \bigvee \{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

since the morphism $\mathcal{U}$ transforms open filters we have

$$\bigvee \{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\} = \bigvee \{u_{v_{\mathcal{U}F}} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

But this representation is the corresponding to the nucleus $v_{\mathcal{U}F}$.

□